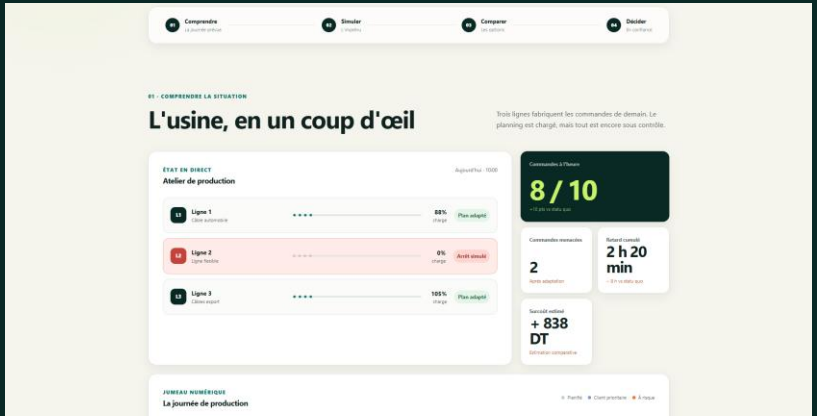
CableTwin

When a line stops, see the consequences before changing production.

The Waze of the cable factory

Working offline prototype · Fictional Tunisian factory · Fully synthetic data



THE PROBLEM

At 10:00, one stoppage exposes 3 orders and 620 minutes of delay

Which orders should be moved, delayed or protected — and what will each choice do to service, throughput and the workshop?

Eligible lines | Durations | Due times | No overlap | Planning horizon

The active schedule becomes infeasible the moment the line stops: two orders overlap the stop, one downstream order is exposed, and every alternative must respect real constraints. The decision owner is the production manager.

LINE 2 UNAVAILABLE

10:00 – 14:00

ORDERS EXPOSED

3

ON TIME WITHOUT ADAPTATION

7 / 10

PROJECTED CUMULATIVE DELAY

620 min

The manager needs a feasible choice, not another dashboard.

THE PRODUCT

CableTwin turns one disruption into three feasible routes

1 Detect the stop

2 Reveal exposed orders and KPI

3 Compare Service · Cost · Stability

4 Preview, approve and record

Like Waze after a road closes:

same destination, different routes and visible trade-offs — the driver still decides.

One incident → three comparable recovery schedules → one traceable human decision

Read-only decision support · No failure prediction · No machine control

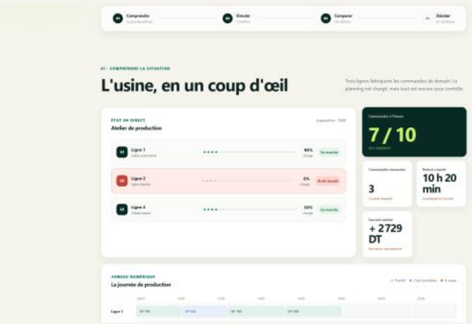
END-TO-END WORKFLOW

One end-to-end loop connects the incident to an auditable decision

01 — Understand

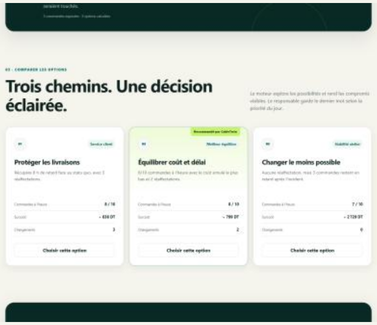
3 lines · 10/10 on time

02 — Simulate



Line 2 stops · 3 exposed

03 — Compare



3 feasible routes · same KPI

04 — Decide

Manager approves · audit at 10:07

The same executable path connects data, incident analysis, planning, Gantt preview, approval, audit and reset.

WORKING PROTOTYPE · MEASURED CONSEQUENCE

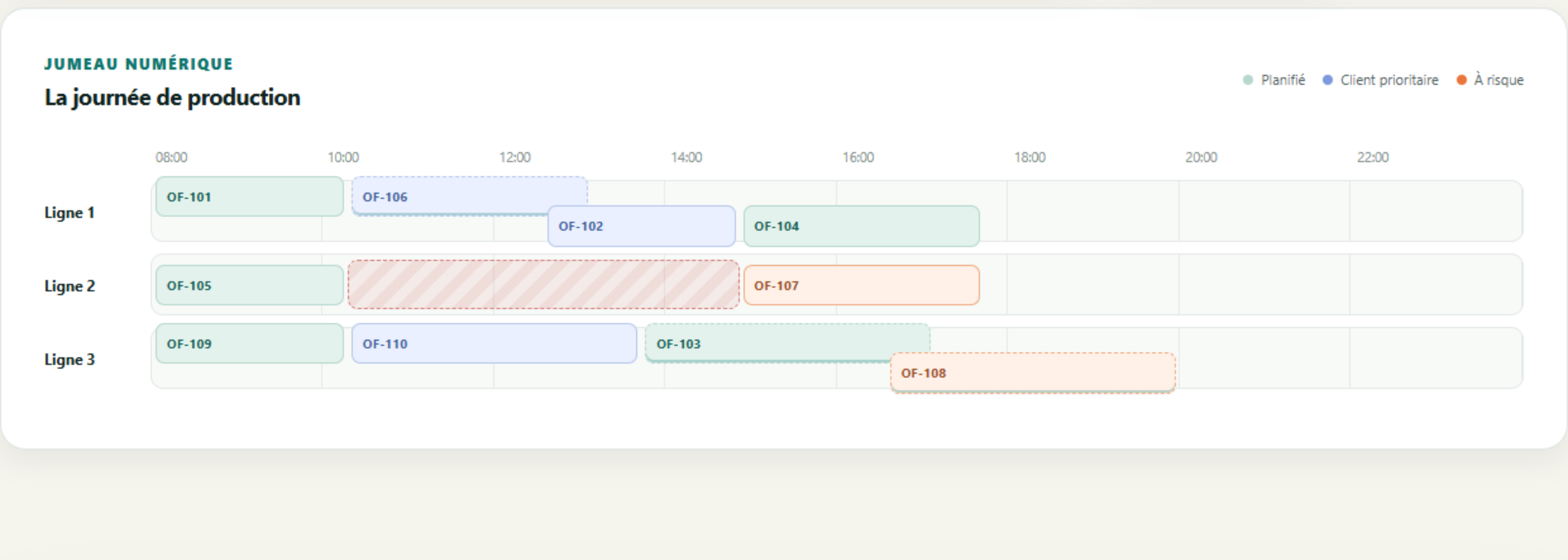
Service cuts projected delay from 620 to 140 minutes

620 → 140 min
projected cumulative delay

188 → 216 km
completed by 18:00 · +14.9%

180 → 30 min
overtime · −83.3%

7/10 → 8/10
orders on time · 3 line moves



Stability
620 min · 188 km · 0 moves

Service
140 min · 216 km · 3 moves

Cost
170 min · 216 km · 2 moves

Fixed synthetic scenario · calculated by the working prototype · not factory ROI

The AI is the planning decision, not the dashboard

LIVE SYMBOLIC PLANNER

Decision variables

line assignment · sequence · start / end

Hard constraints

eligibility · duration · downtime · no overlap · horizon

Business policies

Service · Cost · Stability (lexicographic objectives)

INDEPENDENT BOUNDED ORACLE

17,856 candidate schedules evaluated

10,440 feasible complete schedules

3/3 displayed routes = unique policy optimum

9/9 automated checks pass

A 9/10 route exists, but creates 230 min of total delay; Service chooses 8/10 with only 140 min because priority-weighted lateness matters. Exact for the encoded bounded demo — not factory-scale optimization.

Simulation proves the mechanism; a shadow pilot will prove value

PROVED IN DEMO

Same incident

Same KPI definitions

Reproducible before/after calculations

5–10 historical stoppages

Same information cutoff as the real decision

Planner-validated constraints

CableTwin vs actual historical baseline

Time, delay, output, overtime, moves, usability

PROPOSED SUCCESS GATES — targets to agree, not achieved claims

30% faster assessment · 20% lower delay · 0 hard-constraint violations · ≥70% planner usability · 100% traceability

Ends in Go · Revise · Stop

BUSINESS VIABILITY

Start read-only in one workshop; scale only after measured proof

User: **Production manager / planner** Buyer: **Plant or operations director** Pilot: **4–6 weeks · read-only**

- **Validate process**
- **Map read-only data**
- **Encode workshop constraints**
- **Replay and shadow**
- **Measured go / revise / stop**

Manual / spreadsheet	familiar but hard to compare and trace
ERP · MES · APS	system of record / broad planning
High-fidelity twin	rich but integration-heavy
CableTwin	focused incident layer · 3 explainable routes · human approval

No prediction · no autonomous control · no write-back · no price or ROI claim

READY FOR PHASE 3

CableTwin is ready for Phase 3

25% · Business process & impact	One incident-rescheduling loop · delay, overtime and output measured
20% · Core AI	Constraint planning · 17,856-candidate independent bounded certificate
20% · Working prototype	Offline incident → routes → Gantt → approval → audit → reset · 9/9 checks
20% · Business viability	Named user and buyer · 4–6-week read-only pilot · go/no-go gates
15% · Pitch & clarity	One Waze metaphor · three routes · one human decision

Select CableTwin for the final pitch and see the complete live decision in 90 seconds.